

P8109 Spring 2026 Homework #4

Due on April 18 at 11:59pm

1. Consider sampling from the parent population with density

$$f(x|\theta) = \theta x^{\theta-1}, \quad 0 < x < 1, \theta > 0.$$

Based on a random sample X_1 of size $n = 1$ from this parent population, what is the power of the most powerful (MP) test of

$$H_0 : \theta = 1 \quad \text{versus} \quad H_1 : \theta = 2$$

if

$$P(\text{Type I error}) = \alpha = 0.05?$$

2. Consider sampling from the parent population with density

$$f(x|\theta) = \theta x^{\theta-1}, \quad 0 < x < 1, \theta > 0.$$

If X_1 and X_2 constitute a random sample of size $n = 2$ from this parent population, show that the rejection region of size α for the most powerful test of

$$H_0 : \theta = 1 \quad \text{versus} \quad H_1 : \theta = 2$$

is

$$X_1 X_2 \geq k^*,$$

where

$$k^* - k^* \log k^* = 1 - \alpha.$$

3. Let $Y_1, \dots, Y_n$ be a random sample from the density

$$f(y|\theta) = \frac{1}{2}(1 + \theta y), \quad -1 < y < 1, \theta > -1.$$

- (a) Develop an explicit expression for the form of the MP rejection region R for testing

$$H_0 : \theta = 0 \quad \text{vs.} \quad H_1 : \theta = 1$$

when $P(\text{Type I error}) = \alpha$.

- (b) If $n = 1$ and $\alpha = 0.05$, find the numerical value of the dividing point between the rejection and non-rejection regions for this MP test.

- (c) If, in fact, $\theta = 1$, what is the exact numerical value of the power of this MP test of

$$H_0 : \theta = 0 \quad \text{vs.} \quad H_1 : \theta = 1$$

when $\alpha = 0.05$ and $n = 1$?

4. Suppose $X \sim \text{Poisson}(\lambda)$. A sample of size $n = 10$ is used to test

$$H_0 : \lambda = 0.1 \quad \text{vs.} \quad H_1 : \lambda > 0.1.$$

Construct the randomized MP test of size $\alpha = 0.05$.